

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I S.Blessika(192525251) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Blessika

Assessment Tasks - Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure

uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations. Explain the cloud architecture, services used, security mechanisms, and benefits of the system.

Answer:

Introduction

A Smart Retail Inventory Management System is a cloud-based solution that helps retail stores manage inventory, sales, and demand forecasting efficiently. It connects store devices with centralized cloud services using APIs, ensuring real-time data synchronization and secure communication.

Architecture

- **Clients:** POS systems and dashboards.
- **API Layer:** REST APIs and API Gateway.
- **Cloud Services:** Inventory management, analytics, and forecasting.
- **Cloud Storage:** Product data, transaction logs, analytics datasets.
- **Security:** VPN, Firewalls, Authentication.

Working

1. POS systems record customer purchases.
2. Data is sent to cloud servers through REST APIs.
3. API Gateway manages and secures API requests.
4. Cloud storage stores inventory and transaction data.
5. Platform services analyze sales trends and predict demand.
6. Dashboards display real-time inventory reports.
7. Managers receive alerts for low stock levels.

Cloud Services Used

- **IaaS:** Virtual servers, storage, networking.
- **PaaS:** Analytics and demand forecasting.
- **SaaS:** Inventory management dashboard.

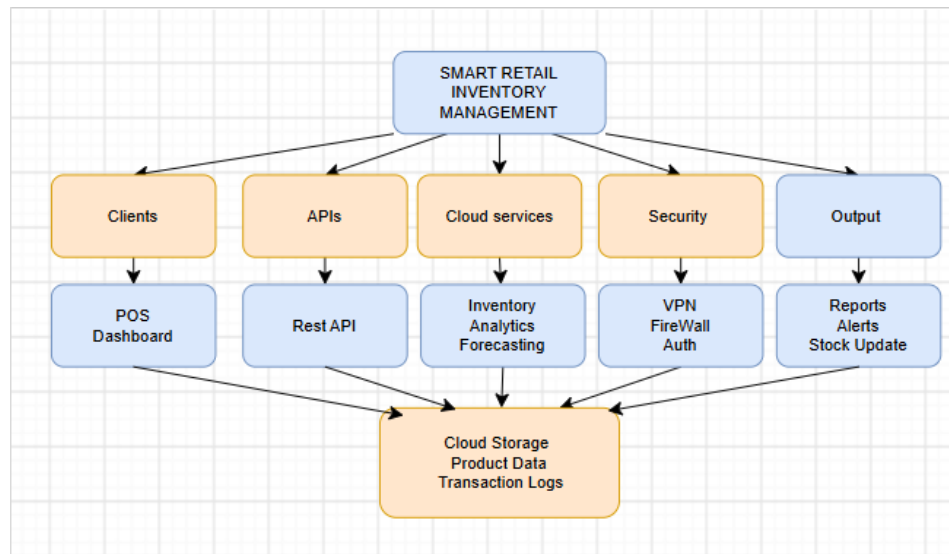
Security Mechanisms

- VPN
- Firewall
- API Gateway
- Authentication and Authorization

Benefits

- Real-time inventory monitoring
- Accurate demand forecasting
- Reduced stock shortages

- Improved business decisions
- High scalability
- Secure communication



2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration. Explain the architecture, cloud services, security mechanisms, and advantages of this system.

Answer:

Introduction

A Cloud-Based Document Collaboration System allows multiple users to create, edit, and share documents online simultaneously. It uses cloud storage and web APIs to synchronize document changes in real time.

Architecture

- **Clients:** Web browsers and mobile devices.
- **Web Application:** User interface for editing documents.
- **Web APIs:** Synchronization between users.
- **Cloud Storage:** Distributed storage with version control.
- **Security:** Encryption and Identity Management.

Working

1. Users log in using web browsers.
2. Documents are opened from cloud storage.
3. Users edit documents simultaneously.
4. Web APIs synchronize all changes instantly.
5. Version control saves document history.
6. Identity management verifies users.

7. Encrypted communication protects data.

Cloud Services Used

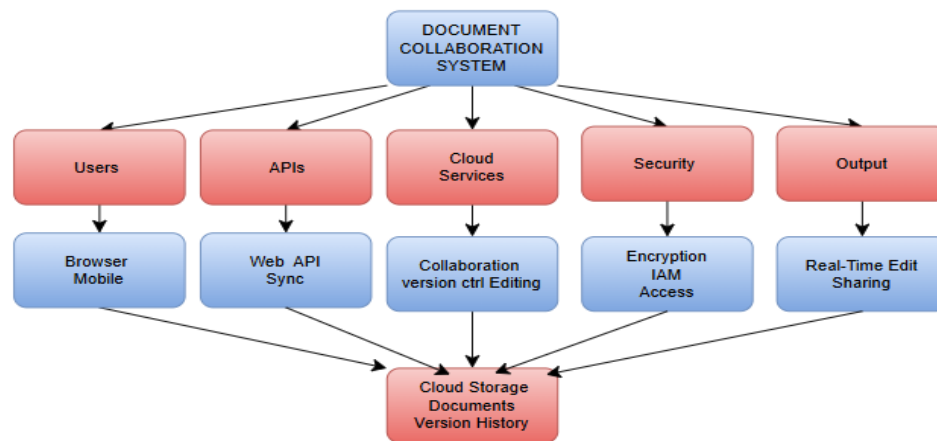
- **SaaS:** Online document editor.
- **PaaS:** Collaboration and synchronization services.
- **Cloud Storage:** Distributed storage.

Security Mechanisms

- Encryption
- Identity and Access Management (IAM)
- User Authentication
- Access Permissions

Advantages

- Real-time collaboration
- Automatic synchronization
- Version control
- Global accessibility
- High availability
- Secure document sharing



3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. Explain the cloud architecture, services used, security mechanisms, and benefits of this platform.

Answer:

Introduction

An Online Food Ordering Platform is a cloud-based application that enables customers to order food through web browsers or mobile apps. It uses cloud infrastructure, REST APIs, and cloud storage for efficient order processing and secure communication.

Architecture

- **Clients:** Web browsers and mobile applications.
- **RESTful APIs:** Communication between frontend and backend.
- **Backend Services:** Order processing and payment management.
- **Cloud Storage:** Menu images, invoices, customer data.
- **Infrastructure:** Virtual Machines and Kubernetes.
- **Security:** HTTPS, Authentication Tokens, RBAC.

Working

1. Customers browse restaurants and menus.
2. Orders are placed through web or mobile apps.
3. REST APIs send requests to backend services.
4. Backend processes orders and payments.
5. Cloud storage stores customer details and invoices.
6. Kubernetes automatically scales services during high traffic.
7. Customers receive real-time order updates.

Cloud Services Used

- **IaaS:** Virtual Machines.
- **PaaS:** Managed Kubernetes.
- **Cloud Storage:** Images, invoices, customer data.
- **REST APIs:** Client-server communication.

Security Mechanisms

- HTTPS
- Authentication Tokens
- Role-Based Access Control (RBAC)
- Secure Cloud Storage

Advantages

- Fast order processing
- Real-time order tracking
- Automatic scalability
- Secure customer information
- High availability
- Improved customer satisfaction

